On the Emergence of Social Hierarchies in a Society

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1 Introduction

Understanding how social hierarchies emerge from initially egalitarian conditions is a fundamental question in the study of complex systems. In particular, agent-based models (ABM) offer a powerful framework to investigate such processes by simulating the behavior and interactions of heterogeneous individuals. These models allow us to uncover emergent patterns and macroscopic dynamics arising from simple local rules, and have been widely used to model social, economic, and biological systems (4; 5).

Social hierarchies are widespread across both human and animal societies. In cities, they can emerge between two distinct groups, such as migrants or tourists versus native residents, or low- versus high-income populations competing for housing, jobs, or influence. They also appear in rivalries as two criminal gangs contesting control over a neighborhood. At the individual level, hierarchies can form spontaneously through repeated interactions, as seen in primate dominance structures (3), workplace dynamics where informal leadership arises, or online platforms where a few users accumulate visibility and authority. These patterns often result from mechanisms such as competition, reinforcement, or memory, and help explain how inequality and stratification emerge even in initially equal settings.

In this project, we aim to model and observe the emergence of social hierarchies among individuals and across groups, starting from an egalitarian initial state. Our approach is based on the seminal work of Bonabeau et al. (1), who proposed a minimal agent-based model in which repeated interactions lead to spontaneous formation of a hierarchical structure. We further explore the extension proposed by Marc Sadurni et al. (2), which introduces two competitive groups into the model, acting as a form of relaxation mechanism to stabilize dynamics.

Our contribution is threefold. First, we implement the original Bonabeau model as well as the two-group extension. Second, we replicate the results from the literature and investigate how various parameters influence the emergence and stability of hierarchies. Third, we propose two novel extensions: one introducing an asymmetrical importance of winning between groups, and another allowing for intra-group interactions.

The report is organized as follows. Section 2 introduces the Bonabeau model and summarizes its key dynamics. Section 3 describes the two-group extension and compares its behavior with the original model. Sections 4 and 5 present our extensions: an asymmetrical importance of winning between groups, and intra-group interactions, respectively. The complete implementation is publicly available on GitHub[1].

[1] https://github.com/CorentinPernot/interactions

2 The Bonabeau Model

2.1 Description

Motivation and Hypothesis

The idea of the Bonabeau model is to start from a simple principle, inspired by bees, and based on a positive feedback mechanism to explain the emergence of social hierarchies. The central assumption is that an individual's probability of winning future encounters increases with each previous victory.

Dynamics of Interactions

Each agent i in the population is associated with a scalar quantity F_i, referred to as its "force" or "fitness", which evolves over time. This quantity can be interpreted as the current physiological condition of the individual. When two individuals i and j meet, the probability that i wins the interaction is given by a sigmoid function:

$$Q_{ij}^{+} = \frac{1}{1 + e^{-\eta(F_i - F_j)}}, \tag{1}$$

where $\eta > 0$ controls the determinism of the outcome: high values of η yield nearly deterministic outcomes favoring the stronger individual, while low values introduce randomness. We have that the probability for agent i to win is equal to the probability for agent j to lose $Q_{ij}^{+} = Q_{ji}^{-} = 1 - Q_{ij}^{-}$.

After each interaction, the winner's force increases by one unit and the loser's force decreases by one unit:

$$\begin{aligned} F_i &\leftarrow F_i + 1 \quad \text{(if } i \text{ wins)} \\ F_j &\leftarrow F_j - 1 \quad \text{(if } j \text{ loses)} \end{aligned}$$

Relaxation and Forgetting Mechanism

To model the fading memory of past interactions, the Bonabeau model incorporates a relaxation process. In the absence of new interactions, an individual's force F_i naturally relaxes toward zero over time. This is governed by the following update rule:

$$F_i \leftarrow F_i - \mu \tanh(F_i), \tag{2}$$

where $\mu > 0$ is the relaxation or forgetting rate. The hyperbolic tangent ensures a linear decay for small F_i and a saturating decay for large magnitudes. This mechanism prevents the indefinite accumulation of dominance and allows for dynamic restructuring of the hierarchy when interactions become sparse.

Quantifying Hierarchy

To evaluate the degree of hierarchy at any point in time, the model uses two complementary indicators:

- **Instantaneous Winning Probability P_i.** This quantity captures the current ability of individual i to dominate others. It is defined as the average probability of i winning a confrontation against any other agent:

$$< P_i >= \frac{1}{N-1} \sum_{j \neq i} \frac{1}{1 + e^{-\eta(F_i - F_j)}}. \tag{3}$$

  It reflects the agent's relative strength at a given moment, and varies as its force F_i evolves through interactions and relaxation. As such, P_i provides a *dynamic* snapshot of the individual's status within the system.

- **Dominance Index X_i.** To capture a more stable and experimentally accessible measure of hierarchy, the model also tracks the ratio of past victories to total encounters:

$$X_i = \frac{D_i}{D_i + S_i}, \tag{4}$$

where D_i and S_i denote the number of wins and losses, respectively. X_i changes only when an interaction occurs, and remains constant otherwise. Hence, it offers a *static*, long-term perspective on an individual's average dominance level.

The two indicators converge asymptotically when agents interact regularly. However, in low-density settings or during periods of inactivity, they may diverge: P_i continues to evolve due to force relaxation, while X_i remains frozen. Therefore, P_i gives a more accurate picture of the current hierarchical structure, while X_i is better suited for summarizing cumulative social performance over time and is computationally easier to access.

To quantify the degree of hierarchy, the model introduces an order parameter:

$$\sigma^2 = \sum_{i=1}^{N} (X_i - 0.5)^2, \tag{5}$$

which vanishes in the egalitarian phase and increases as hierarchy emerges.

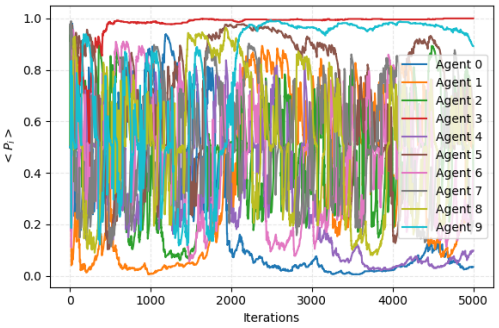


Figure 1.a. $\rho = 0.1$

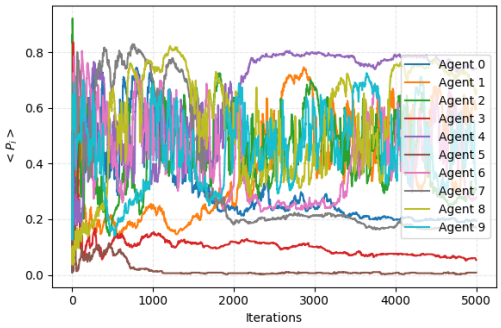


Figure 1.b. $\rho = 0.2$

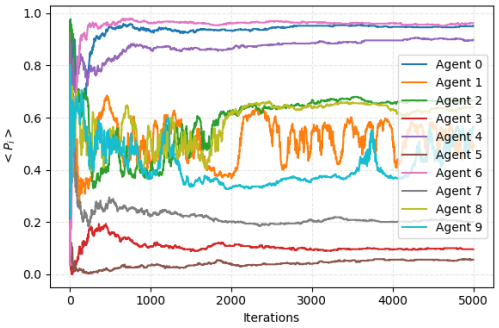


Figure 1.c. $\rho = 0.5$

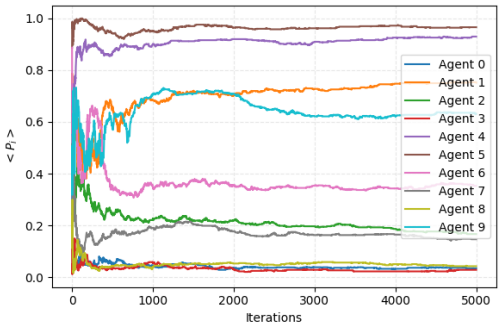


Figure 1.d. $\rho = 0.8$

Figure 1: Evolution of the instantaneous probability that individual i wins in a random fight $< P_i >$ for different values of the density ρ and for 10 individuals. The simulations were run for 5000 iterations with $\eta = 0.5$ and $\mu = 0.1$.

Monte Carlo Simulation

As in the original paper (1), we aim to visualize the emergence of hierarchies from an initially non-hierarchical society. To this end, we perform Monte Carlo simulations on a two-dimensional grid of size $L \times L$, where N agents execute random walks. At each iteration, individuals move to a new site. If two agents occupy the same site, they engage in a fight and update their respective "fitness" levels.

The original model assumes that no more than two agents can share the same site. In our version, if more than two agents end up on the same site, we randomly select two of them to fight, while the others remain passive "spectators."

Finally, at the end of each iteration, the fitness of all individuals is relaxed towards zero.

2.2 Influence of the parameters

Density ρ

We ran simulations with $N = 500$ individuals over 5000 time steps. Hierarchies are observed to emerge when the density $\rho := N/L^2$, which is controlled by adjusting the grid size L, exceeds a certain threshold. In other words, when ρ is too low, no clear hierarchy develops. This is because at low densities, interactions between agents are too infrequent, allowing the relaxation term to dominate the dynamics.

We display the evolution of the instantaneous winning probability P_i over time. In these simulations, we fixed the parameters to $\eta = 0.5$ and $\mu = 0.1$, following the original setup in (1). We tested various values of the density parameter ρ, specifically 0.1, 0.2, 0.5, and 0.8 (see Figure 1).

In Figure 1.a, we clearly observe that when $\rho \approx 0.1$, there is no stable asymptotic profile and the dynamics appear erratic. As ρ increases, the asymptotic behavior becomes more stable and is reached more quickly. In Figures 1.c and 1.d, fluctuations are rare, and Bonabeau et al. (1) referred to this as the "ordered" phase.

Furthermore, as shown in Figure 12 (Appendix), higher densities lead to more pronounced hierarchies. When the density is sufficiently high, the competition term dominates the relaxation term, reinforcing inequalities and leading to a hierarchical society. Conversely, when the relaxation term dominates, the society remains egalitarian, with only a few leaders and losers, and their fitness remains moderate compared to those in a hierarchical structure.

Indeed, the extreme values of fitness become significantly higher, and the society more polarized, when the density is high (see Figure 2.b). For instance, the fitness of the most successful agents can reach values as high as 1500 when $\rho = 0.5$, while remaining bounded by 150 when $\rho = 0.1$ (Figure 2.a).

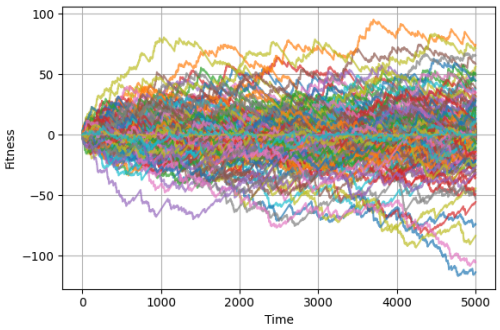


Figure 2.a. $\rho = 0.1$

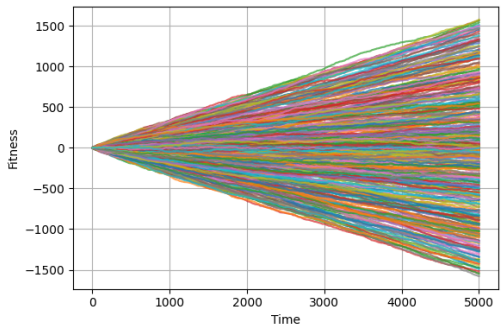


Figure 2.b. $\rho = 0.5$

Figure 2: Evolution of the fitness of all agents for different densities ρ.

Parameter Influence and Phase Transition in the Mean-Field Model

In this work, we did not reproduce the simulations exploring the role of the parameters η and μ. However, following Bonabeau et al. (1), we recall that η controls the sensitivity of the outcome of a fight to the fitness difference between agents: it plays the role of the inverse temperature. A higher η amplifies the dominance of stronger agents, thus fostering hierarchy formation. Conversely, μ controls the rate of relaxation of fitness: increasing μ tends to homogenize the population and counteract hierarchical differentiation.

To gain a theoretical understanding of these dynamics, Bonabeau et al. (1) proposed a mean-field approach. They derived a simplified evolution equation for the variance of the fitness distribution, capturing the balance between competition (amplified by η and the agent density ρ) and relaxation (driven by μ). This leads to a phase diagram characterizing a transition from a disordered (egalitarian) to an ordered (hierarchical) phase. We reproduce this phase diagram in Figure 3 to visualize the critical frontier separating these regimes.

The nature of the bifurcation depends explicitly on η:

- For $\eta < \frac{2\mu}{(1+1/N)\rho}$: **subcritical bifurcation (discontinuous)** hierarchy emerges abruptly, and bistability is possible between flat and hierarchical profiles.
- For $\eta > \frac{2\mu}{(1+1/N)\rho}$: **supercritical bifurcation (continuous)** hierarchy develops smoothly as p increases beyond the threshold.
- For $\eta = \frac{2\mu}{(1+1/N)\rho}$: a **tricritical point** occurs, separating the two regimes.

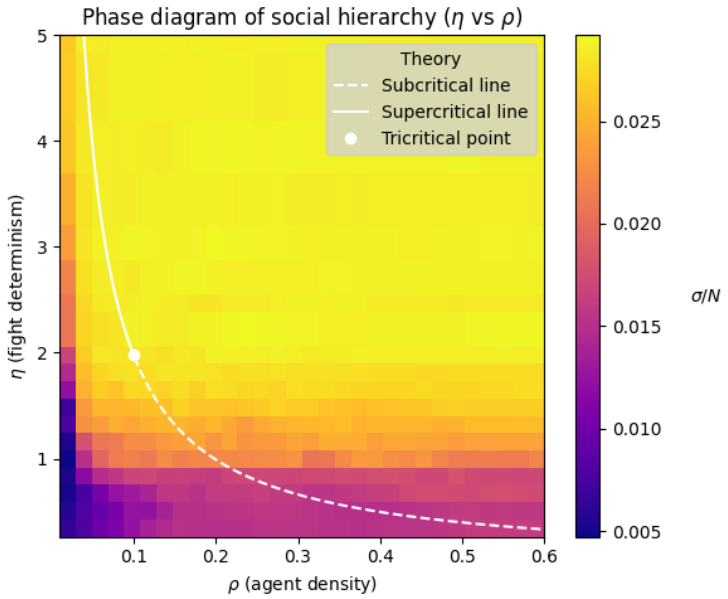


Figure 3: Phase diagram of social hierarchy in the (η, ρ) plane. The color indicates the normalized order parameter σ/N, measuring the degree of hierarchy. The solid and dashed white lines represent the theoretical bifurcation line predicted by the mean-field approximation. The tricritical point separates the subcritical (discontinuous) and supercritical (continuous) transition regimes. Parameters: $N = 500$, $\mu = 0.1$, $Iterations = 5000$

Numerical Phase Diagram and Validation

To validate the theoretical predictions, we constructed a numerical phase diagram by systematically varying both the density ρ and the determinism parameter η. For each pair (ρ, η), we computed the normalized order parameter σ/N at equilibrium, which quantifies the degree of hierarchy in the population. The resulting diagram is shown in Figure 3.

We observe a sharp transition between a disordered egalitarian phase (low σ/N, shown in dark purple) and an ordered hierarchical phase (high σ/N, shown in yellow). This transition occurs along a critical curve that matches well with the mean-field theoretical prediction:

$$\rho_c = \frac{2\mu}{(1 + 1/N)\eta},$$

with $\mu = 0.1$, $N = 500$ and 5000 iterations, in our simulations. The theoretical bifurcation line (solid and dashed white curves) separates the two regimes, and we also identify the tricritical point (white dot), which marks the boundary between subcritical and supercritical bifurcations.

In the subcritical regime ($\eta < 2/(1 + 1/N)$), the transition is abrupt and consistent with a discontinuous shift from egalitarian to hierarchical profiles. In contrast, in the supercritical regime ($\eta > 2/(1 + 1/N)$), the order parameter increases smoothly as ρ grows, reflecting a continuous emergence of social stratification. Our simulations reproduce both behaviors, providing numerical evidence for the qualitative structure predicted by the mean-field theory developed by Bonabeau et al. (1).

3 The Model with two groups

3.1 Motivation and links with Bonabeau

The original Bonabeau model relies on two key mechanisms: a reinforcement process, where agents gain or lose fitness through interactions, and a relaxation process that gradually reduces fitness in the absence of activity. The emergence of a hierarchical structure depends on the balance between these two forces, which is itself strongly influenced by the agent density ρ.

The model introduced in (2) removes the relaxation term and instead introduces a second group of agents. Interactions are now limited to pairs of individuals from opposing groups. This change brings a new form of regulation: since agents can only gain or lose fitness through inter-group competition, the internal dynamics of each group become indirectly constrained.

A notable difference with the original model is the way interactions depend on fitness. While Bonabeau's model uses absolute fitness values, here the authors compare normalized fitness within each group. This normalization reflects the relative prestige of individuals among their peers and avoids scenarios where one group could end up absorbing all the fitness in the system. Indeed, without normalization, repeated victories could lead to the collapse of one groups influence.

Another consequence of the interaction rules is that total fitness remains constant over time. Unlike in Bonabeau's formulation, where dominant agents can accumulate arbitrarily high fitness, here the system behaves more like a closed economy, with fitness simply redistributed across individuals.

3.2 Formalization

Formally the dynamics of the system, described in (2) can be written as follows:

We consider a system composed of two distinct populations: group A with N_A agents and group B with N_B agents, evolving on a two-dimensional lattice of size $L \times L$. As before, at each time step, agents perform independent random walks on the grid and intercation occurs when a site is occupied simultaneously by one agent from group A and one from group B.

Each agent i in group A and agent j in group B is associated with a fitness value $F_i^A(t)$ and $F_j^B(t)$, respectively. Before computing the probability of winning, these values are normalized within each group to capture relative prestige:

$$\tilde{F}_i^A(t) = \frac{F_i^A(t) - F_{\min}^A(t)}{F_{\max}^A(t) - F_{\min}^A(t)}, \quad \tilde{F}_j^B(t) = \frac{F_j^B(t) - F_{\min}^B(t)}{F_{\max}^B(t) - F_{\min}^B(t)}.$$

The probability that agent i from group A defeats agent j from group B is given by a logistic function of the normalized fitness difference:

$$P_{ij}(t) = \frac{1}{1 + \exp\left[\eta(\tilde{F}_j^B(t) - \tilde{F}_i^A(t))\right]},$$

where $\eta > 0$ controls the determinism of the interactions. High values of η favor the stronger agent, while low values introduce more randomness.

The fitness update depends on the outcome of the fight. If i wins, the fitness is updated as follows:

$$F_i^A(t + \Delta t) = F_i^A(t) + xF_j^B(t), \quad F_j^B(t + \Delta t) = F_j^B(t) - xF_j^B(t).$$

Conversely, if i loses:

$$F_i^A(t + \Delta t) = F_i^A(t) - xF_i^A(t), \quad F_j^B(t + \Delta t) = F_j^B(t) + xF_i^A(t).$$

The exchange parameter $x \in (0, 1)$ controls how much fitness is transferred during each interaction. By construction, the total fitness of the system is preserved at all times:

$$\phi(t) = \sum_{i=1}^{N_A} F_i^A(t) + \sum_{j=1}^{N_B} F_j^B(t) = 1000.$$

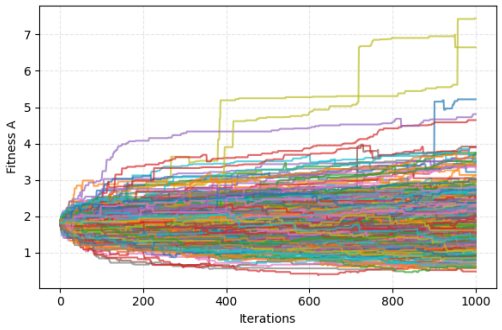


Figure 4.a. Group A

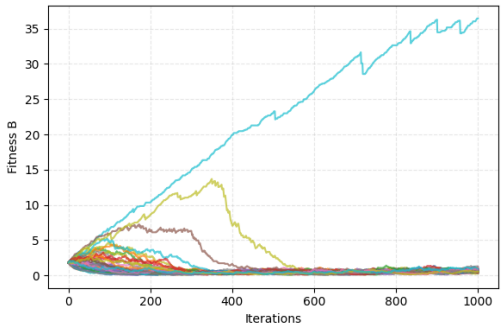


Figure 4.b. Group B

Figure 4: $\eta = 5$. Parameters: $N_A = 500,\ N_B = 50,\ L = 25,\ x = 0.05,\ Iter = 1000$.

3.3 Results

Emergence of a leader

We reproduced the experimental setup from (2) with the following parameters: $N_A = 500$, $N_B = 50$, $L = 25$, and a total fitness fixed at 1000. However, we slightly increased the exchange rate to $x = 0.05$ in order to accelerate convergence towards a stationary regime. This modification, which introduces more stochasticity into the system, was suggested in the appendix of the original paper.

As illustrated in Figure 4, the emergence of a leader is much more pronounced within the minority group B. While group A exhibits limited dispersion and relatively modest increases in fitness, group B quickly sees the rise of a dominant agent whose fitness grows significantly over time. This asymmetry highlights the effect of group size on the dynamics of hierarchy formation: a smaller group is more susceptible to rapid stratification.

We will further quantify this phenomenon using Lorenz curves and Gini coefficients in the next section.

Stationnary regime

Moreover, we observe that the stationary regime is reached after approximately 20,000 iterations (Figure 5.b), which is faster than what was reported in the original paper. This acceleration is due to the increased exchange rate x, which acts as a catalyst by amplifying the effects of each interaction. However, this comes at the cost of higher variability within the stationary regime, as reflected by the colored fluctuations and intermittent peaks observed in the plot.

This behavior suggests a trade-off between convergence speed and long-term stability: while increasing x allows the system to settle more rapidly into a hierarchical configuration, it also amplifies noise and transient dominance effects, making the stationary state more volatile and potentially less robust.

The Impact of η: Anarchy, Monarchy, or Oligarchy?

The parameter η plays a key role in regulating the degree of stochasticity in the system. Low values of η correspond to highly random outcomes, while high values make interactions more deterministic and favor the emergence of hierarchy.

When $\eta = 1$, the outcome of interactions is almost entirely random, and no clear leader emerges (Figure 5.a). The fitness values fluctuate without structure, and the system fails to reach a stationary regime. This regime resembles a form of social anarchy, where no agent gains sustained advantage over others.

At intermediate values such as $\eta = 5$, we observe the emergence of a dominant agent whose fitness grows steadily over time, eventually stabilizing around a high value (Figure 5.b). The rest of the population sees their fitness reduced to negligible levels. This configuration mirrors a monarchy scenario, where a single leader concentrates all the power and influence.

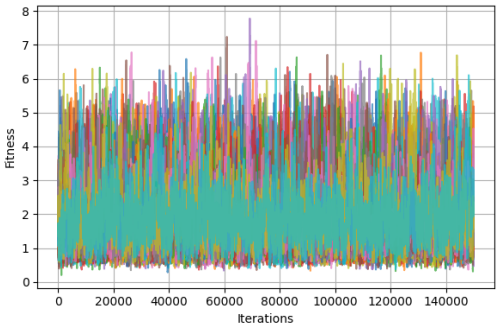


Figure 5.a. $\eta = 1$, no emergence of a leader, no stationnary regime

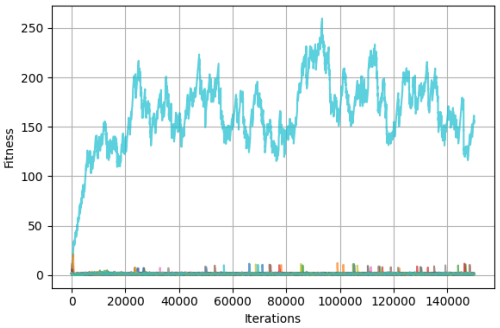


Figure 5.b. $\eta = 5$, emergence of one leader, stationnary regime reached

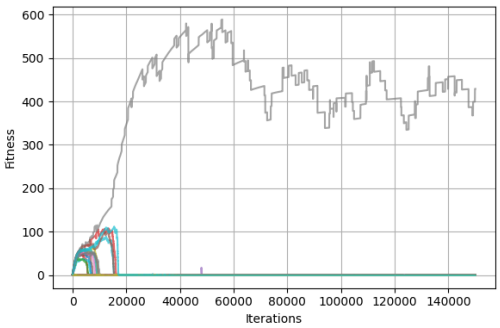


Figure 5.c. $\eta = 10$, emergence of several leaders, but stationnary regime reached only for one

Figure 5: Stationnary regimes for the minority group B for different values of η. Parameters: $N_A = 500,\ N_B = 50,\ L = 25,\ x = 0.05,\ Iterations = 150000$

Interestingly, when η is further increased (e.g., $\eta = 10$), the system initially allows several agents to rise simultaneously during the early iterations (Figure 5.c). However, in the long run, only one of them persists and stabilizes in the stationary regime, while others fade out. This transient coexistence of leaders evokes a form of oligarchy or unstable multi-leadership before a final winner takes over.

These results highlight how η, by controlling the determinism of interactions, acts as a bifurcation parameter that governs the structure of the emergent hierarchyranging from disordered egalitarianism to sharply stratified dominance.

Lorenz Curves and Gini Coefficients

To assess the impact of group size imbalance on inequality, we computed Lorenz curves and Gini indices for two configurations with a fixed total population $N_T = 550$, using parameters $L = 25$, $x = 0.05$, and $\eta = 5$, as in (2).

In the imbalanced setting ($N_A = 500, N_B = 50$, Figure 6.a), group B exhibits a high level of inequality (Gini = 0.624), while group A remains more egalitarian (Gini = 0.283). In the more balanced case ($N_A = 350, N_B = 200$, Figure 6.b), inequality within group B drops significantly (Gini = 0.346), and the overall population becomes more homogeneous (Gini = 0.297 vs. 0.336).

These results confirm that small groups are more vulnerable to hierarchical concentration, with a few individuals quickly dominating the rest. A more balanced group composition mitigates this effect and promotes a flatter fitness distribution.

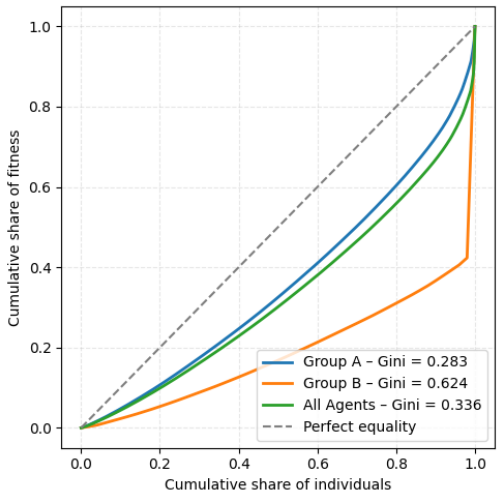


Figure 6.a. $(N_A, N_B) = (500, 50)$

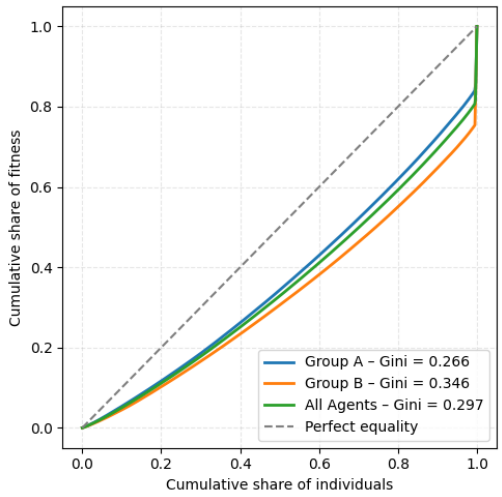


Figure 6.b. $(N_A, N_B) = (350, 200)$

Figure 6: Lorenz curves

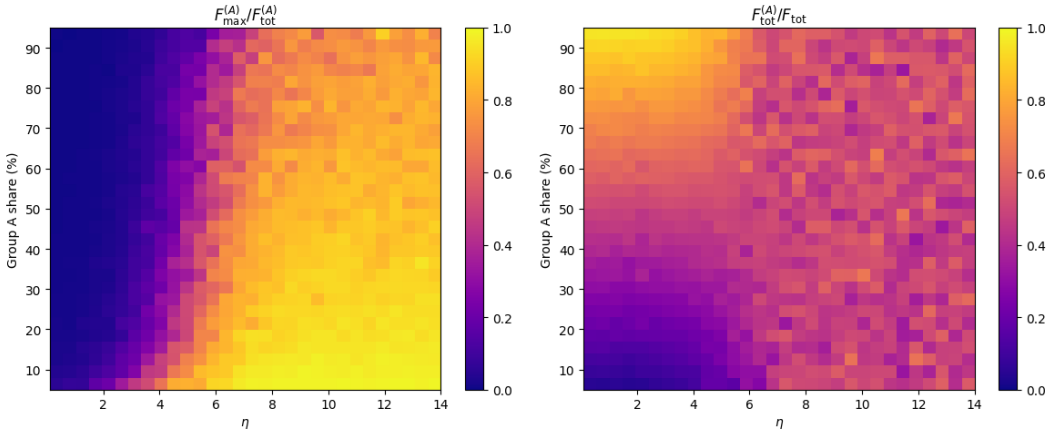


Figure 7: Heatmaps for two quantities related to group A: (left) $F^{(A)}_{\max}/F^{(A)}_{\text{tot}}$, indicating the degree of internal hierarchy; (right) $F^{(A)}_{\text{tot}}/F_{\text{tot}}$, representing the share of global fitness captured by group A. Parameters: $N_T = 1000$, $x = 0.05$, $F_{\text{tot}} = 1000$, *Iterations* $= 40000$.

Phase Diagram

We present in Figure 7 the phase diagram quantifying the influence of group proportion and determinism on hierarchy formation and fitness allocation. The x-axis corresponds to the parameter η, while the y-axis represents the percentage share of group A in the total population.

The left panel shows the normalized maximum fitness within group A, $F^{(A)}_{\max}/F^{(A)}_{\text{tot}}$, which reflects the strength of hierarchy in that group. We observe that the emergence of a leader is favored when the group is small. For instance, when group A represents only 10 or 30% of the population, a dominant agent quickly emerges. The transition is particularly sharp: for a 50% group share, no hierarchy appears at $\eta = 4$, while a clear leader is present at $\eta = 7$. The higher the value of η, the more pronounced the hierarchy becomes.

The right panel displays the relative share of global fitness captured by group A, $F^{(A)}_{\text{tot}}/F_{\text{tot}}$. This offers a potential explanation for the differences in asymptotic fitness observed in Figures 5.b and 5.c. When a group represents only 10% of the population ($\approx N_{group} = 50$ out of 550), its share of the total fitness is about 20% for $\eta = 5$ and rises to around 50% for $\eta = 10$ according to 7. These values

match the empirical fitness levels seen in the simulations: around 200 for $\eta = 5$ and close to 450 for $\eta = 10$, out of a total of 1000.

Furthermore, we note that for $\eta > 5$, the total fitness tends to be equally distributed between the two groups, regardless of their sizes. Below that threshold, however, fitness distribution becomes more asymmetric and follows a gradient that depends on group proportion and interaction randomness.

These results consolidate our earlier findings and offer a macroscopic view of how group size and determinism jointly shape social stratification.

3.4 Comparison with Bonabeau

The two-group model introduced by Sadurni et al. (2) reinterprets Bonabeaus mechanism (1) through a different lens. While Bonabeaus framework hinges on the interplay between reinforcement and relaxation, the two-group model removes the relaxation term and replaces it with constrained inter-group interactions. This structural shift decouples the dynamics from interaction frequency and anchors them instead in group composition and relative prestige. Normalizing fitness within each group prevents unbounded divergence and acts as a stabilizing feedback, but also suppresses absolute dominance. Unlike Bonabeau's open system, where fitness accumulates over time, the Sadurni model conserves total fitness, turning hierarchy formation into a redistributive process. From a dynamical perspective, dissipation is replaced by constraint, enabling faster convergence but increasing sensitivity to asymmetries. The presence of distinct groups also introduces phenomena absent in the original model, such as instability driven by group imbalance or transient multi-leadership regimes. Ultimately, the two-group formulation is not just an extension but a conceptual shiftfrom self-organization through decay to stratification through structured competition in a closed system.

4 Asymetrical winner importance

We have seen that the previous model was able to produce the emergence of one or several leaders and achieves then desired properties in the modelling of population dynamics. The previous update of the fitness characteristic was:

$$F_i^A(t + \Delta t) = F_i^A(t) + xF_j^B(t), \quad F_j^B(t + \Delta t) = F_j^B(t) - xF_j^B(t).$$

in the case of a victory of i and with a change of sign conversely. This corresponds to a symmetrical transfer, no matter who wins, the same quantity of fitness will be exchanged at the end of the fight. An interesting extension of the model is to study the case where the exchanged quantity depends on the winner of the fight. To this aim, we now fix $(x_a, x_b) \in (0, 1)^2$ and the update rules are:

$$F_i^A(t + \Delta t) = F_i^A(t) + x_aF_j^B(t), \quad F_j^B(t + \Delta t) = F_j^B(t) - x_aF_j^B(t).$$

when i wins and:

$$F_i^A(t + \Delta t) = F_i^A(t) - x_bF_i^A(t), \quad F_j^B(t + \Delta t) = F_j^B(t) + x_bF_i^A(t).$$

when he loses. One might notice that this extension is still conservative in the sense that in all cases, we have:

$$F_i^A(t + \Delta t) + F_j^B(t + \Delta t) = F_i^A(t) + F_j^B(t).$$

The two factors x_a and x_b can be interpreted as the importance of a victory for a given group. For example, consider the extreme case (not allowed in our setup) where $x_a = 0$ and $x_b = 1$. When agent i wins, their fitness remains unchanged. Conversely, if agent j wins, the fitness of i drops to zero, while j gains all of i's fitness. This extreme scenario illustrates that the outcome of a fight can affect the system differently depending on the group to which the winner belongs. In this section, we numerically investigate the behavior of such a process.

In Figure 8, the extreme settings $x_a = 0.01$ and $x_b = 0.90$ confirm the intuition we had about this new feature. The significant changes in the fitness of group A are due to the large coefficient x_b: at each loss, an agent from group A loses 90% of its fitness, leading to erratic behavior. On the other hand, the small coefficient x_a causes the fitness curve of group B to remain relatively smooth.

Reversing the extreme setting, *i.e.*, setting $x_a = 0.99$ and $x_b = 0.01$, yields another interesting phenomenon. In this case, a leader slowly emerges in group A, while in group B, each loss causes

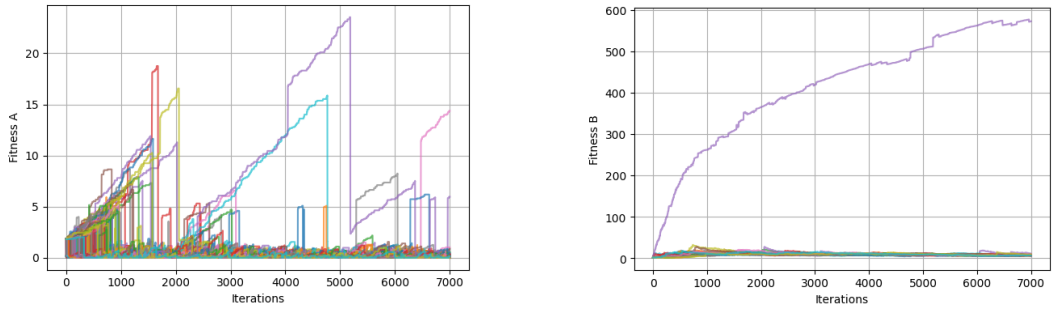


Figure 8: Fitness of group A and B when using $x_a = 0.01$ and $x_b = 0.90$

individuals to lose 99% of their fitness, effectively "resetting" the process and allowing a new leader to appear.

Naturally, one may want to explore intermediate settings, where the fitness dynamics are less abrupt. As mentioned earlier, this new feature could help model more complex social structures and may represent a step toward capturing different patterns of hierarchical evolution.

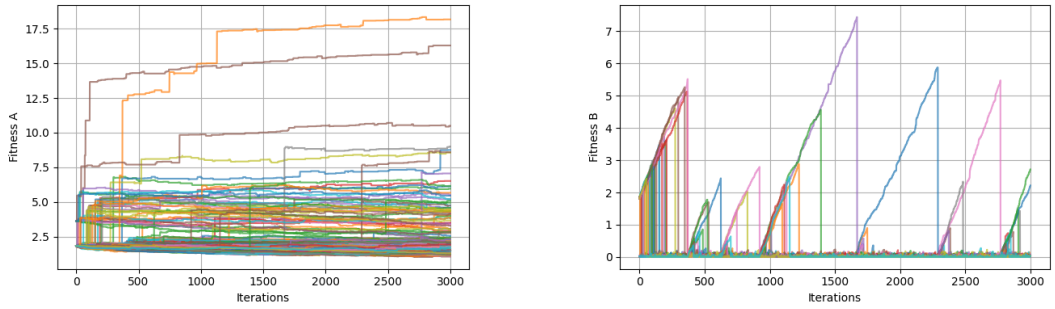


Figure 9: Fitness of group A and B when using $x_a = 0.99$ and $x_b = 0.01$

This improvement is a first step but might be sophisticated, and future research could study time-dependent rates, *i.e.*, rates of the form $x_a(t)$ and $x_b(t)$. It could model a time-dependent society (weather evolution, resource depletion, etc...) that is nowadays under extensive research.

We also mention two very interesting future research tracks. As we mentioned before, the fitness process of group B in Figure 9 is a succession of different leaders and represents a less stable society. It might be of interest to study the average lifetime of a leader to characterize a typical temporal order of magnitude of the system.

The last track that we think is of interest is to determine the expression of the fitness of group B in Figure 8. Due to the extreme coefficient, the curves do not (almost) vary across the different experiments, suggesting that the variance of such a process tends to zero and the average process might be found explicitly.

5 Extension: Adding Intra-Group Interactions

5.1 Motivation

The original model by Sadurni et al. (2) only allows inter-group interactions, neglecting the internal competition often present in real-world systems. In political parties, firms, or social classes, individuals also compete within their own group for power or visibility. To reflect this, we extend the model by introducing intra-group interactions, controlled by a new exchange parameter x_{intra}, alongside the existing inter-group parameter x_{inter}.

At each iteration, if multiple agents share a grid cell, we probabilistically choose to trigger an intra- or inter-group fight (30% vs. 70%), provided both are possible. Intra-group fights follow the same rules: the winner, determined by normalized fitness, receives a fraction x_{intra} of the loser's fitness. This extension preserves total fitness but enables hierarchies to form both within and across groups. We now investigate the effect of this addition through comparative simulations.

5.2 Effect of intra-group competition on group dynamics

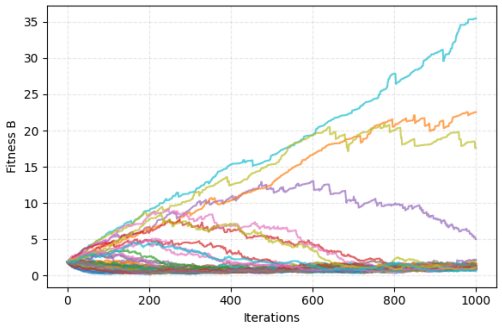


Figure 10.a. $x_{\text{inter}} = 0.05 > x_{\text{intra}} = 0.02$: external competition dominates. A few agents steadily accumulate fitness by defeating group A.

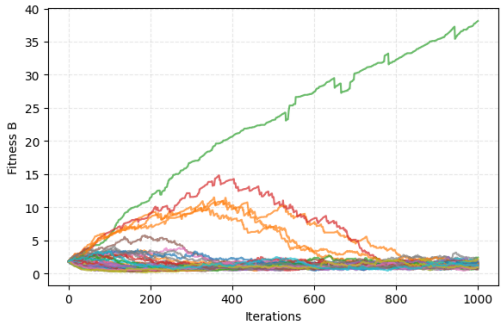


Figure 10.b. $x_{\text{inter}} = x_{\text{intra}} = 0.05$: volatile regime. Early winners rise quickly with steep fitness gains, but often fall just as sharply.

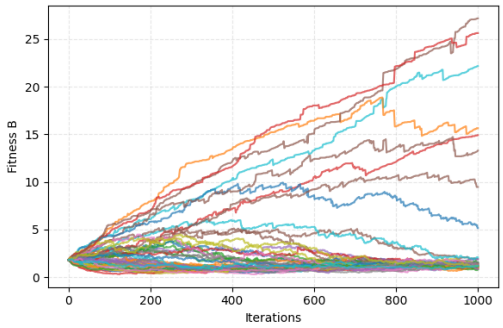


Figure 10.c. $x_{\text{intra}} = 0.05 > x_{\text{inter}} = 0.02$: internal rivalry.

Figure 10: Fitness trajectories for the minority group B under different intra/inter-group interaction regimes. Parameters: $N_A = 550,\ N_B = 50,\ L = 25,\ \eta = 5,$ Iterations $= 1000$.

We now focus on the behavior of both groups in a strongly asymmetric setting ($N_A = 550$, $N_B = 50$). Figure 10 shows the evolution of individual fitness within minority group B under three regimes: (i) dominant inter-group interactions, (ii) balanced intra/inter-group competition, and (iii) dominant intra-group interactions.

For group B, when inter-group interactions dominate ($x_{\text{inter}} = 0.05 > x_{\text{intra}} = 0.02$), only a few agents manage to steadily increase their fitness (Figure 10.a). These survivors benefit from repeated victories over the majority group. The rest of the group stagnates or declines, highlighting the structural disadvantage of being outnumbered. In the balanced case ($x_{\text{inter}} = x_{\text{intra}} = 0.05$), early winners rise sharply but often lose ground later (Figure 10.b). This regime is more volatile, with frequent leadership turnover and unstable trajectories. When intra-group interactions dominate ($x_{\text{intra}} = 0.05 > x_{\text{inter}} = 0.02$), internal competition leads to rapid polarization (Figure 10.c). A few individuals accumulate high fitness early on and maintain it, while others are weakened from within. Here, hierarchy stems more from internal domination than external success. For group A, the fitness trajectories are more stable overall.

When inter-group interactions dominate, growth is steady and slopes are moderate, reflecting gradual hierarchy formation driven by external success. In the balanced case, the slopes become steeper for a few individuals, as internal competition amplifies fitness divergence. When intra-group interactions

dominate, the slopes are overall lower, suggesting that excessive internal rivalry may slow down the emergence of dominant agents.

Overall, these results show that intra-group rivalry shapes not only the structure of inequality, but also the mechanisms behind it. While inter-group competition tends to reward external success, strong intra-group interactions can slow down the formation of dominant individuals. Similar levels of inequality may emerge under different regimes, but the underlying dynamics remain distinct.

6 Limits and Further Improvements

The Bonabeau model and its two-group extension are simple but useful to study how hierarchies emerge. However, they make several strong assumptions. First, the idea of having only two fixed groups is not always realistic. In real societies, group boundaries are often fuzzy or changing. This makes it hard to isolate the effect of one parameter. In this context, the phase diagram is helpful. It gives a global view of how the system behaves depending on the key parameters.

Another limit is the absence of intra-group interactions. In real life, people also compete within their group. There is rivalry, jealousy, or informal leadership. Our extension tries to fix this by allowing intra-group fights, with their own interaction rate. We also studied an asymmetric case, where the importance of winning depends on the group. This makes the model more flexible.

One more limitation is that agent movement is completely random. In reality, individuals often have strategies. They may avoid strong opponents, or seek visibility. This spatial behavior could be improved in future versions.

We think about three possible improvements. First, we could allow the interaction rates $x_a(t)$ and $x_b(t)$ to vary over time. This would model societies where competition evolves, for example due to external shocks. Second, we could explore a more collective setting. Agents could coordinate their fights instead of acting alone. Finally, we could add a third group to generalize to more complex situations. This would allow us to study alliances or multi-group competition.

7 Conclusion

In this project, we explored the emergence of social hierarchies through agent-based models, from Bonabeau's original formulation to a two-group extension. We replicated key results and proposed two improvements: asymmetric interaction rates and intra-group competition. These additions allowed us to model more realistic dynamics and revealed new phenomena such as transient leadership. We also highlighted the value of phase diagrams as a powerful tool to visualize how key parameters shape the system's behavior, helping to identify transitions between egalitarian, oligarchic, and hierarchical regimes. While simple, the model offers rich dynamics and can be further extended by adding group-level strategies, temporal variations, or multi-group structures. This opens promising directions for future work on collective social dynamics.

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8 Appendix

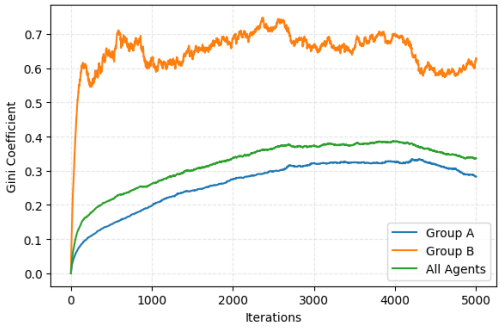


Figure 11.a. $(N_A, N_B) = (500, 50)$

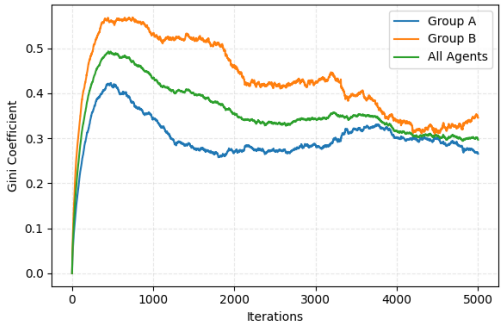


Figure 11.b. $(N_A, N_B) = (350, 200)$

Figure 11: Gini coefficients

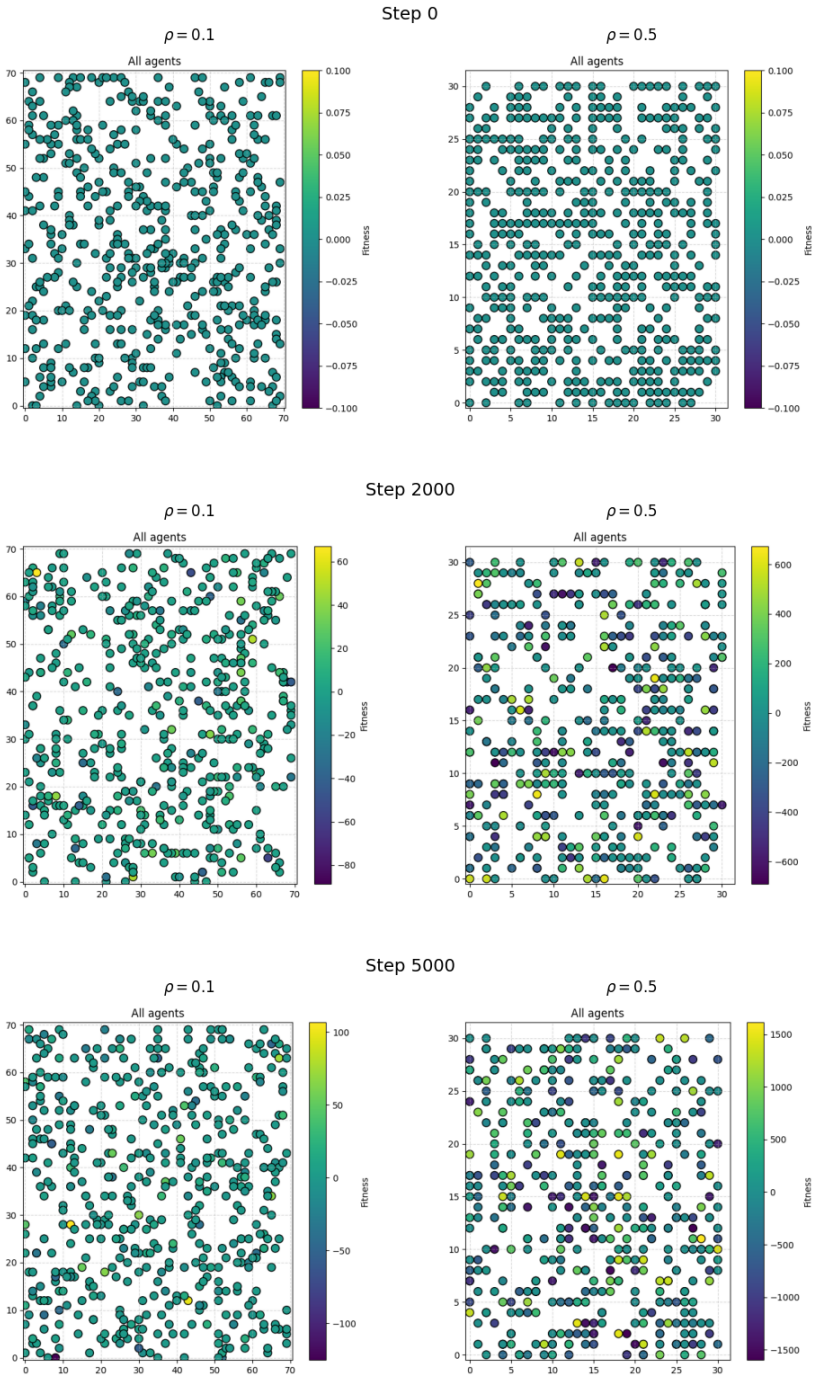


Figure 12: Evolution of fitness across the entire society for two density levels ρ. When $\rho = 0.1$, no hierarchy emerges and inequalities remain moderate. In contrast, when $\rho = 0.5$, the society becomes hierarchical and inequalities grow significantly in both directions.